

数学物理方法作业

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第十四章

6. 因为方程非齐次项 $bx(l-x)$ 只是 x 的函数, 设方程有只关于 x 的特解 $v(x)$ 。

即设:

$$u(x, t) = v(x) + w(x, t)$$

则特解 $v(x)$ 满足:

$$\frac{\partial^2 v}{\partial t^2} - a^2 \frac{\partial^2 v}{\partial x^2} = -a^2 \frac{\partial^2 v}{\partial x^2} = bx(l-x)$$
$$v(0) = v(l) = 0$$

积分得一个解:

$$v(x) = \frac{b}{12a}x^4 - \frac{bl}{6a^2}x^3$$

通解 $w(x, t)$ 满足:

$$\frac{\partial^2 w}{\partial t^2} - a^2 \frac{\partial^2 w}{\partial x^2} = 0$$
$$w(0, t) = w(l, t) = 0$$
$$w(x, 0) = -v(x), \quad \left. \frac{\partial w}{\partial t} \right|_{t=0} = 0$$

设解的形式为 $w(x, t) = X(x)T(t)$, 代入方程整理得:

$$X(x)T''(t) = a^2 X''(x)T(t)$$

因为 $X(x)T(t) \neq 0$, 上式两边同时除以 $a^2 X(x)T(t)$ 得:

$$\frac{T''(t)}{a^2 T(t)} = \frac{X''(x)}{X(x)}$$

设它们都等于同一个非零常数 $-\lambda$, 即:

$$\frac{T''(t)}{a^2 T(t)} = \frac{X''(x)}{X(x)} = -\lambda$$

得到两个方程:

$$X''(x) + \lambda X(x) = 0, \quad T''(t) + a^2 \lambda T(t) = 0$$

关于 x 的本征值问题为:

$$X''(x) + \lambda X(x) = 0, \quad X(0) = 0, \quad X'(l) = 0$$

其通解为:

$$X(x) = A \sin \sqrt{\lambda}x + B \cos \sqrt{\lambda}x$$

代入边界条件得:

$$B = 0, \quad \sqrt{\lambda}l = n\pi$$

本征值和本征函数为:

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n = \sin \frac{n\pi}{l}x$$

代入关于 t 的方程, 解得:

$$T_n = C_n \sin \frac{n\pi}{l}at + D_n \cos \frac{n\pi}{l}at$$

一般解为:

$$w(x, t) = \sum_{n=0}^{\infty} \left[C_n \sin \frac{n\pi}{l}at + D_n \cos \frac{n\pi}{l}at \right] \sin \frac{2n+1}{2l}\pi x$$

代入初始条件 $w(x, 0) = -v(x)$ 得:

$$\sum_{n=0}^{\infty} D_n \sin \frac{n\pi}{2l}x = -\frac{b}{12a}x^4 + \frac{bl}{6a^2}x^3$$

注意到 D_n 为 Fourier 展开的系数, 解得:

$$D_n = \frac{2}{l} \int_0^l \left(-\frac{b}{12a}x^4 + \frac{bl}{6a^2}x^3 \right) \sin \frac{n\pi}{2l}x \, dx = \begin{cases} 0, & n = 2k \\ -\frac{8bl^4}{a^2\pi^5 n^5}, & n = 2k+1 \end{cases}$$

代入 $\frac{\partial w}{\partial t} \Big|_{t=0} = 0$ 得 $C_n = 0$ 。则通解 $w(x, t)$ 为:

$$w(x, t) = -\frac{8bl^4}{a^2\pi^5} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^5} \cos \frac{2k+1}{l}\pi at \sin \frac{2k+1}{l}\pi x$$

故原方程的解为:

$$u(x, t) = \frac{b}{12a}x^4 - \frac{bl}{6a^2}x^3 - \frac{8bl^4}{a^2\pi^5} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^5} \cos \frac{2k+1}{l}\pi at \sin \frac{2k+1}{l}\pi x$$

7.(1) 用单本征函数展开法, 相应的齐次方程为:

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

设解的形式为 $u(x, t) = X(x)Y(y)$, 代入方程整理得:

$$X(x)Y''(y) = X''(x)Y(y)$$

因为 $X(x)Y(y) \neq 0$, 上式两边同时除以 $X(x)Y(y)$ 得:

$$\frac{Y''(y)}{Y(y)} = \frac{X''(x)}{X(x)}$$

设它们都等于同一个非零常数 $-\lambda$, 即:

$$\frac{Y''(y)}{Y(y)} = \frac{X''(x)}{X(x)} = -\lambda$$

得到两个方程:

$$X''(x) + \lambda X(x) = 0, \quad T''(t) + a^2 \lambda Y(y) = 0$$

关于 x 的本征值问题为:

$$X''(x) + \lambda X(x) = 0, \quad X(0) = 0, \quad X(l) = 0$$

其通解为:

$$X(x) = A \sin \sqrt{\lambda} x + B \cos \sqrt{\lambda} x$$

代入边界条件得:

$$B = 0, \quad \sqrt{\lambda} a = n\pi$$

本征值和本征函数为:

$$\lambda_n = \left(\frac{n\pi}{a}\right)^2, \quad X_n = \sin \frac{n\pi}{a} x$$

设解为:

$$u(x, y) = \sum_{n=0}^{\infty} Y_n(y) \sin \frac{n\pi}{a} x$$

将非齐次项 -2 用本征函数展开:

$$-2 = \sum_{n=0}^{\infty} C_n \sin \frac{n\pi}{a} x$$

解得:

$$C_n = \frac{2}{a} \int_0^a \left(-2 \sin \frac{n\pi}{a} x\right) dx = \frac{4}{n\pi} [(-1)^n - 1]$$

则关于 y 的方程和边界条件为:

$$Y_n''(y) - \left(\frac{n\pi}{a}\right)^2 Y_n(y) = \frac{4}{n\pi} [(-1)^n - 1]$$

$$Y_n\left(\frac{b}{2}\right) = Y_n\left(-\frac{b}{2}\right) = 0$$

解得:

$$Y_n(y) = \begin{cases} 0, & n = 2k \\ \frac{8a^2}{\pi^3 n^3} \left(1 - \frac{\cosh \frac{n\pi y}{a}}{\cosh \frac{n\pi b}{2a}}\right), & n = 2k + 1 \end{cases}$$

则原方程的解为:

$$\begin{aligned} u(x, t) &= \sum_{n=0}^{\infty} X_n(x) Y_n(y) \\ &= \frac{8a^2}{\pi^3} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^3} \left(1 - \frac{\cosh \frac{2k+1}{a} \pi y}{\cosh \frac{2k+1}{2a} \pi b}\right) \sin \frac{2k+1}{a} \pi x \end{aligned}$$

8. 由题意, 杆的纵振动 $u(x, t)$ 所满足的方程、边界条件和初始条件为:

$$\begin{aligned}\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} &= A \sin \omega t \\ u(0, t) &= 0, \quad \left. \frac{\partial u}{\partial x} \right|_{x=l} = \frac{A}{ES} \sin \omega t \\ u(x, 0) &= 0, \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = 0\end{aligned}$$

设方程有如下形式的特解:

$$v(x, t) = f(x) \sin \omega t$$

即设:

$$u(x, t) = v(x, t) + w(x, t) = f(x) \sin \omega t + w(x, t)$$

则特解 $v(x, t)$ 满足:

$$\frac{\partial^2 v}{\partial t^2} - a^2 \frac{\partial^2 v}{\partial x^2} = A \sin \omega t$$

化简得:

$$a^2 f''(x) + \omega^2 f(x) + A = 0$$

代入

$$v(0, t) = 0, \quad \left. \frac{\partial v}{\partial x} \right|_{x=l} = \frac{A}{ES} \sin \omega t$$

解得:

$$v(x, t) = \frac{Aa \sin\left(\frac{\omega x}{a}\right)}{ES\omega \cos\left(\frac{\omega l}{a}\right)} \sin \omega t$$

通解 $w(x, t)$ 满足:

$$\begin{aligned}\frac{\partial^2 w}{\partial t^2} - a^2 \frac{\partial^2 w}{\partial x^2} &= 0 \\ w(0, t) &= 0, \quad \left. \frac{\partial w}{\partial x} \right|_{x=l} = 0 \\ w(x, 0) &= 0, \quad \left. \frac{\partial w}{\partial t} \right|_{t=0} = - \left. \frac{\partial v}{\partial t} \right|_{t=0} = - \frac{Aa \sin\left(\frac{\omega x}{a}\right)}{ES \cos\left(\frac{\omega l}{a}\right)}\end{aligned}$$

解得:

$$w(x, t) = \frac{4a\omega}{ES\pi a} \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} \cdot \frac{\sin \frac{2k+1}{2l} \pi x}{\left(\frac{\omega}{a}\right)^2 - \left(\frac{2k+1}{2l} \pi\right)^2} \sin \frac{2n+1}{2l} \pi at$$

则原方程的解为:

$$u(x, t) = \frac{Aa \sin\left(\frac{\omega x}{a}\right)}{ES\omega \cos\left(\frac{\omega l}{a}\right)} \sin \omega t + \frac{4a\omega}{ES\pi a} \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} \cdot \frac{\sin \frac{2k+1}{2l} \pi x}{\left(\frac{\omega}{a}\right)^2 - \left(\frac{2k+1}{2l} \pi\right)^2} \sin \frac{2n+1}{2l} \pi at$$

若发生共振，设此时 ω 为：

$$\omega_0 = \frac{2m+1}{2l}\pi a$$

则将 $w(x, t)$ 中 $k = m$ 项与 $v(x, t)$ 合并，再取 $\omega \rightarrow \omega_0$ 的极限，由 L'Hospital 法则得：

$$\begin{aligned} u(x, t) = & \frac{6Al(-1)^m}{ES(2m+1)^2\pi^2} \sin \frac{2m+1}{2l}\pi x \sin \frac{2m+1}{2l}\pi at \\ & + \frac{2A(-1)^{m+1}}{ES(2m+1)\pi} \left(x \cos \frac{2m+1}{2l}\pi x \sin \frac{2m+1}{2l}\pi at + at \sin \frac{2m+1}{2l}\pi x \cos \frac{2m+1}{2l}\pi at \right) \\ & + \frac{8Al(2m+1)}{ES\pi^2} \sum_{k=0, k \neq m}^{\infty} \frac{(-1)^k}{(2k+1)[(2m+1)^2 - (2k+1)^2]} \sin \frac{2k+1}{2l}\pi x \sin \frac{2k+1}{2l}\pi at \end{aligned}$$